Diffeomorphic Learning for Nonlinear Classification

Revising the Mathematical Flow, Geometry, & Numerical Results

Final Presentation

April 9, 2026

Agenda

1. Motivation and Introduction
2. Mathematical Formulation
3. The Core Geometric Trick: Adding a Dimension
4. Computational Challenges
5. Numerical Tours and Results
6. Implementation & Conclusion

Motivation

Core objective: Learn a smooth global deformation that makes difficult, non-linearly separable classification problems simpler for a final linear model.

Why this matters:

- Many datasets are topologically bound and not linearly separable in their original geometry.
- Deep black-box nonlinear models can be accurate but less interpretable.
- Diffeomorphic learning provides an explicit, invertible geometric transformation governed by physical energy constraints.

Classifier form:

$$x \mapsto F(\phi(x); \theta).$$

Introduction: Geometric Learning View

- ϕ is a **diffeomorphism**: smooth, invertible, and topology-preserving.
- The model moves training samples through a smooth, time-dependent flow (velocity field) to improve separability without tearing the space.
- A simple terminal classifier (e.g. logistic regression) is subsequently applied.

Design principle (The Trade-Off):

- Maximize final prediction accuracy.
- Minimize the complexity (i.e., the physical kinetic energy) of the topological deformation.

Original Cost Function

The objective balances deformation effort, model complexity, and data fit:

$$G(\phi, \theta) = \underbrace{D(\mathrm{id}, \phi)^2}_{\text{Deformation Cost}} + \underbrace{\lambda g(\theta)}_{\text{Regularization}} + \underbrace{\frac{1}{\sigma^2}\Gamma\big(F(\cdot;\theta),\, \phi(1)\cdot T_0\big)}_{\text{Terminal Loss}}$$

- $D(\mathrm{id}, \phi)^2 = \inf_v \int_0^1 \|v(t)\|_V^2 \, dt$: In Riemannian geometry, distance is the shortest path (an L^1 length). However, minimizing this L^2 action (which physically mirrors **Kinetic Energy** $\propto \int v^2$) mathematically guarantees the exact same shortest path, but forces a smooth, constant-speed flow.
- $g(\theta)$: Standard regularization (e.g., L^2) on terminal classifier parameters θ.
- Γ: Cross-entropy data loss that is evaluated **strictly** and only at the final time $t = 1$, assessing how linearly separable the points $\phi(1) \cdot x$ have become.

Kernels and the Function Space (V)

The velocity field $v(t, x)$ at any time t lives in a **Reproducing Kernel Hilbert Space (RKHS)** defined by a kernel $K(x, y)$.

Choices of Kernels matter theoretically:

- **Gaussian RBF:** Smooth (C^∞) transformations.
- **Matérn Kernels:** Highly valuable theoretically as they are equivalent to Sobolev spaces, allowing control over exact differentiability classes (C^k).
- **Graph-Based Kernels:** Used if the data resides on strict grids or defined graphs (breaking isotropic symmetries).

Affine Transformations: A Theoretical Necessity

The Limitation of pure RKHS:

- RKHS kernels (like Gaussians) vanish at infinity. This means the model struggles to perform simple translations and rigid global rotations.

The Fix (Section 4.3):

- We explicitly construct an extended vector space: $\widehat{v}(t, \cdot) = g(t, \cdot) + v(t, \cdot)$, where g belongs to a Lie algebra (e.g., Affine transforms).
- **Crucial Result:** This breaks the rotational invariance of scalar kernels and massively increases the topological expressivity of the model. It is a structural necessity.

How to Optimize? The Kernel Trick & Control Points

Optimizing an infinite-dimensional flow $v(t)$ is impossible numerically.

The RKHS Reduction: Because the end-loss only depends on the trajectories of the N sample points, the Representer theorem guarantees the optimal velocity takes the finite form:

$$v(t, \cdot) = \sum_{k=1}^{N} K(\cdot, z_k(t)) a_k(t)$$

We have transformed an infinite-dimensional problem into finding **finite, time-dependent control vectors** $a_k(t)$ attached to the traveling points $z_k(t)$.

The Complete Reduced Optimal Control Problem

When we include the Affine transformation $g(t,x) = A(t)x + b(t)$, the total velocity field becomes:

$$\widehat{v}(t,\cdot) = g(t,\cdot) + \sum_{k=1}^{N} K(\cdot, z_k(t))\, a_k(t)$$

Plugging this back into our original cost formulation yields our final **computable** objective:

$$\mathcal{E}(a,g,\theta) = \underbrace{\int_0^1 \|g(t)\|_{\mathfrak{a}}^2\, dt}_{\text{Affine Energy}} + \underbrace{\int_0^1 \sum_{k,l} a_k(t)^T K(z_k(t), z_l(t)) a_l(t)\, dt}_{\text{RKHS Energy}} + \lambda g(\theta) + \frac{1}{\sigma^2}\Gamma$$

The updated flow constraints:

$$\dot{z}_k(t) = \widehat{v}(t, z_k(t)), \quad z_k(0) = x_k$$

This completely reduces the continuous setting to an exact **finite-dimensional optimal control problem** over variables a, g, and θ.

Pontryagin Maximum Principle: Why Applicable Here

We have exactly the PMP setting:

- State equation (ODE) driven by continuous controls $a(t)$.
- Integral running cost + terminal loss evaluated at $t = 1$.
- Smooth dynamics and differentiable objective.

Hence, first-order optimality can be characterized via a **Hamiltonian mechanics** system coupling the forward positions with a backward co-state $p(t)$.

Role in practice: PMP mechanics naturally provide the exact **Backpropagation-Through-Time (BPTT)** equations needed to iteratively extract gradients.

Applying PMP: Continuous Optimality System

Define Hamiltonian (control form):

$$H_a(p, z) = \langle p, f(z, a) \rangle - L(z, a),$$

with the drift defined by the RKHS kernel:

$$f_k(z, a) = \sum_l K(z_k, z_l) a_l.$$

PMP conditions (The physics of the flow):

$$\dot{z} = \partial_p H_a \quad \text{(Data flows forward)}, \qquad \dot{p} = -\partial_z H_a \quad \text{(Error flows backward)}.$$

Boundary condition (Terminal gradient at $t = 1$):

$$p(1) = \frac{1}{\sigma^2} \partial_z \Gamma\big(F(\cdot; \theta), T(z(1))\big).$$

Discretized Final Formula (Used Numerically)

Euler discretization ($t = 0, \ldots, T-1$):

$$z(t+1) = z(t) + \frac{1}{T}\,\partial_p H_a(t).$$

Adjoint recursion (BPTT):

$$p(t-1) = p(t) + \frac{1}{T}\,\partial_z H_a(t), \quad p(T-1) = \frac{1}{\sigma^2}\partial_z \Gamma.$$

Gradient form used in updates (for Adam/CG):

$$u_k(t) = \sum_l K\big(z_k(t), z_l(t)\big)\big(p_l(t) - 2a_l(t)\big).$$

Overcoming the Topology Barrier

The Problem:

- Diffeomorphisms are strictly topology-preserving.
- In 1D, this strictly means **preserving monotonicity** (a line cannot fold over itself). In N-D, it means points maintain their neighborhood structure without tearing or gluing space.
- If Class A (inner circle) is completely surrounded by Class B (outer ring), **no strictly 2D diffeomorphism can separate them** without tearing the space!

The Theoretical Solution (Lifting / The Dummy Dimension):

- Adding a dummy dimension is one structural solution proposed within this framework to resolve the topological restriction.
- We augment our data with an extra coordinate: $x \to (x, 0)$.
- In this $(d+1)$-dimensional space, the topological barrier is broken. The velocity field simply lifts Class A **over** or **under** Class B in the dummy dimension, flattening them out perfectly for logistic regression.

Scalability and Limits

Method	Typical Complexity	Practical impact
Diffeomorphic Learning	$\mathcal{O}(TdN^2)$ / iter	Main bottleneck: N^2 kernel coupling prevents scaling natively.
Logistic Regression	$\mathcal{O}(NdC)$ / iter	Very fast; strong on near-linear data.
Kernel SVM	$\mathcal{O}(N^3)$	Strong nonlinear baseline but expensive at scale.
MLP	$\mathcal{O}(Ndh)$ / epoch	Scales well linearly with minibatching.

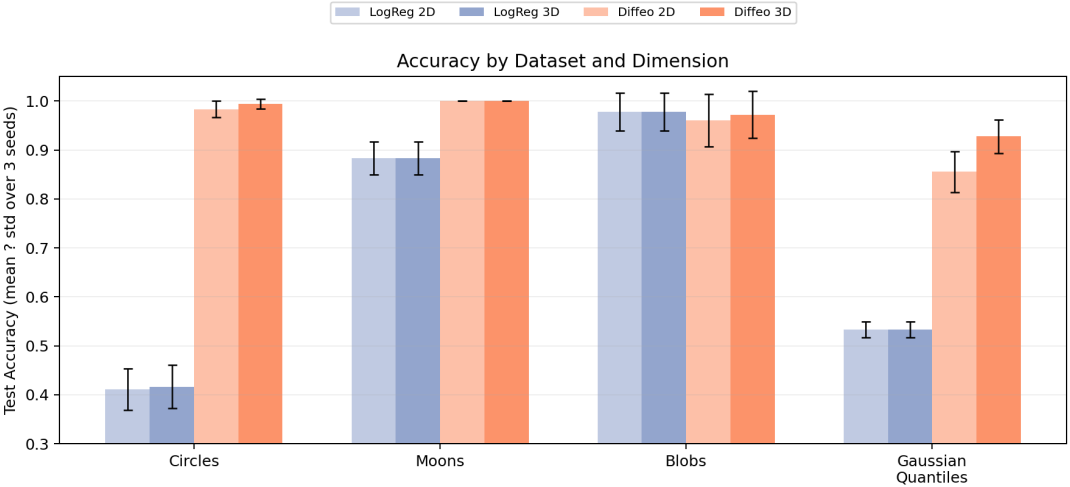
Proposed speedups to bypass N^2 coupling:

- **Control points** ($n \ll N$): Represent flow with sparse landmarks instead of sample points.
- **Mini-batch data term**: Made possible once the control representation is decoupled from N, enabling stochastic gradient descent.
- **Parametric Models** : Replace the exact RKHS representation with a neural-net-like parametric form (e.g., $v(t,x) = \sigma(A(t)x + b(t))$), mapped over finite dimensions.

Experimental Setup

- **Datasets evaluated**:
  - **Circles**: Concentric structures evaluating strict topological inclusion constraints.
  - **Moons**: Intertwined geometries assessing highly non-linear boundary separation capabilities.
  - **Blobs**: Linearly separated gaussian target clusters.
  - **Gaussian Quantiles**: Strongly nested, dense concentric groupings testing complex boundary extraction.
- **Framework Variants**: 2D baseline versus 3D Lifting strategy (employing a dummy dimension initialized with Gaussian perturbation $\sigma = 0.01$).
- **Evaluation Environment**: Metrics reported as the test classification accuracy over 3 invariant initialized seed states.

Performance: 2D vs 3D Across Datasets



Notice the massive accuracy spike on Gaussian quantiles explicitly because of 3D Lifting!

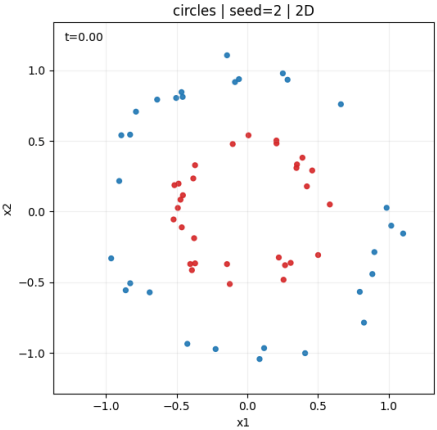
Results Analysis (Final Accuracy Criterion)

Dataset	LogReg 2D	LogReg 3D	Diffeo 2D	Diffeo 3D	Δ Diffeo
Circles	0.411	0.417	0.983	0.994	**+0.011**
Moons	0.883	0.883	1.000	1.000	+0.000
Blobs	0.978	0.978	0.961	0.972	+0.011
Quantiles	0.533	0.533	0.856	0.928	**+0.072**

- **Circles**: Represents a bounded topological barrier. The algorithm leverages 3D lifting to perfectly decouple the locked ring without local minima penalties.
- **Moons**: Devoid of strict inner boundaries. The native 2D continuous flow unbends the intertwined distribution seamlessly, achieving absolute precision accuracy universally.
- **Blobs**: Natively linearly separable configurations. Diffeomorphic routing performs similarly to native linear limits across configurations with standard negligible deviation.
- **Gaussian Quantiles**: The ultimate test for topological lifting, holding distinct intersecting rings. 3D projection mathematically acts as the essential variable enabling internal boundaries to elevate over exterior clusters resulting in the definitive 7.2% accuracy climb.

Results Analysis: Circles Dataset (2D Flow)

Before



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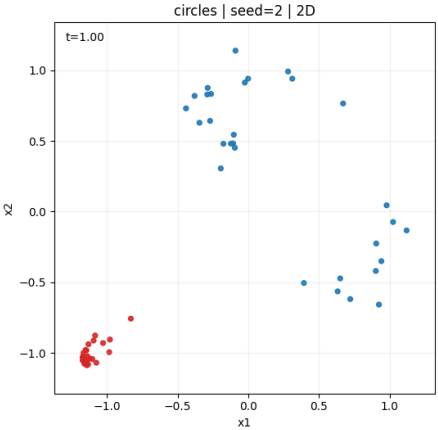
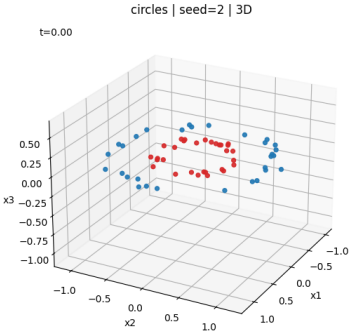


Native 2D deformations on the concentric Circles target. Notice the topological restriction limits complete

Results Analysis: Circles Dataset (3D Lifting)

Before



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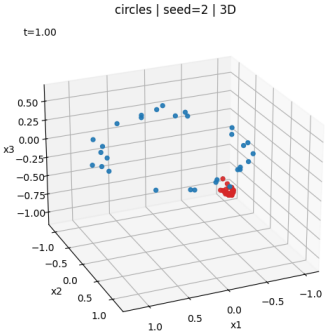
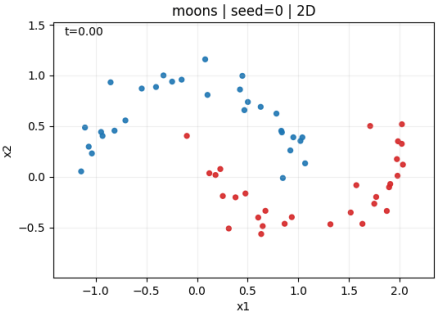


By injecting a 3D dummy dimension, the model correctly separates the locked ring over the surrounding context, reaching near 100% metric convergence.

Results Analysis: Moons Dataset (2D Flow)

Before



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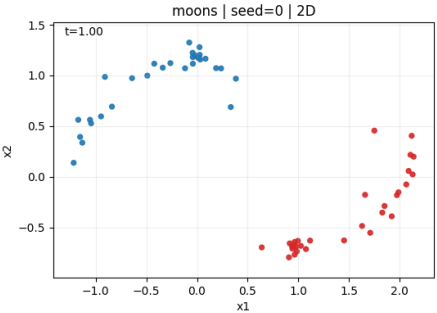
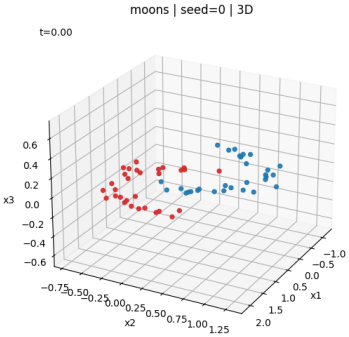


2D spatial deformations successfully unbending the intertwined Moons distributions since they are not bounded

Results Analysis: Moons Dataset (3D Lifting)

Before



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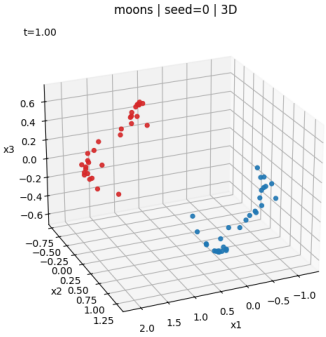
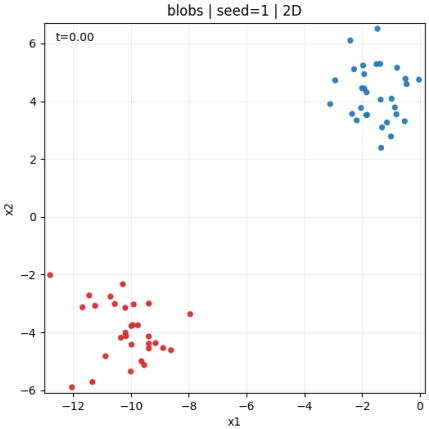


3D Lifting results for the Moons dataset; minimal dimensional gain is noted since the 2D plane flow already achieved full accuracy.

Results Analysis: Blobs Dataset (2D Flow)

Before



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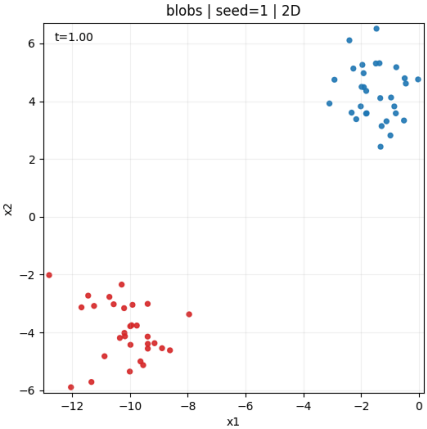
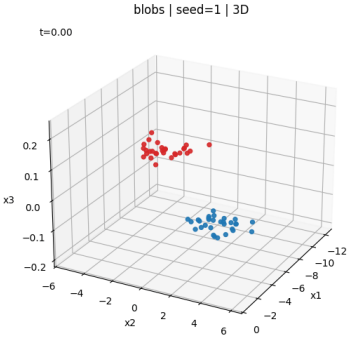


Basic alignment of linearly separable Gaussian Blobs via learned 2D mapping.

Results Analysis: Blobs Dataset (3D Lifting)

Before



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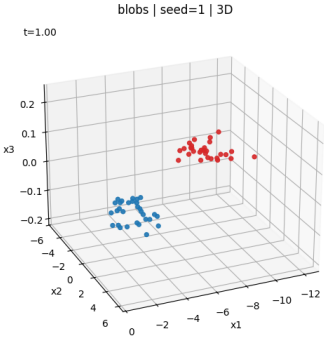
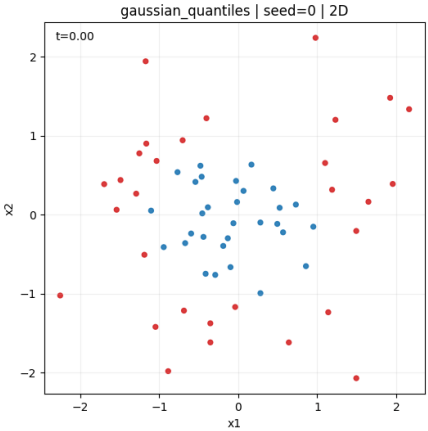


Dummy coordinate lifting over previously separated Blobs; Diffeomorphic models collapse seamlessly to simple affine transforms here.

Results Analysis: Gaussian Quantiles (2D Flow)

Before



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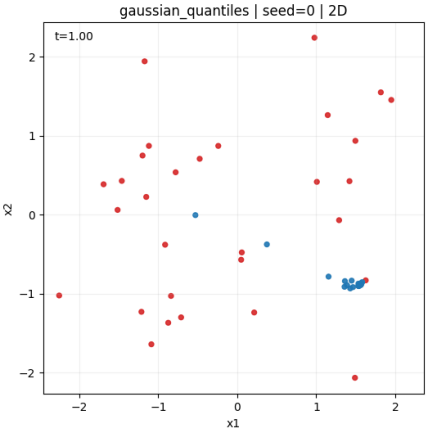
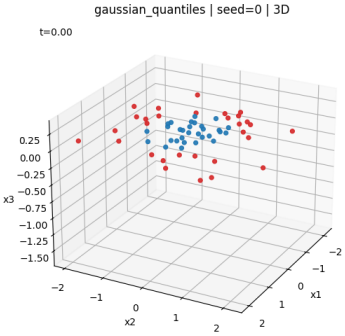


Struggling to cluster interlaced target segments inside the Quantiles dataset natively within the 2D bounding

Results Analysis: Gaussian Quantiles (3D Lifting)

Before



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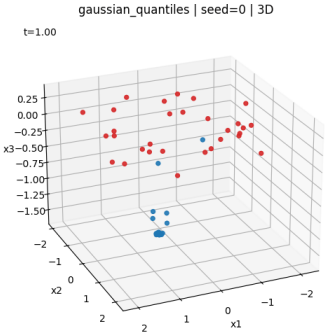
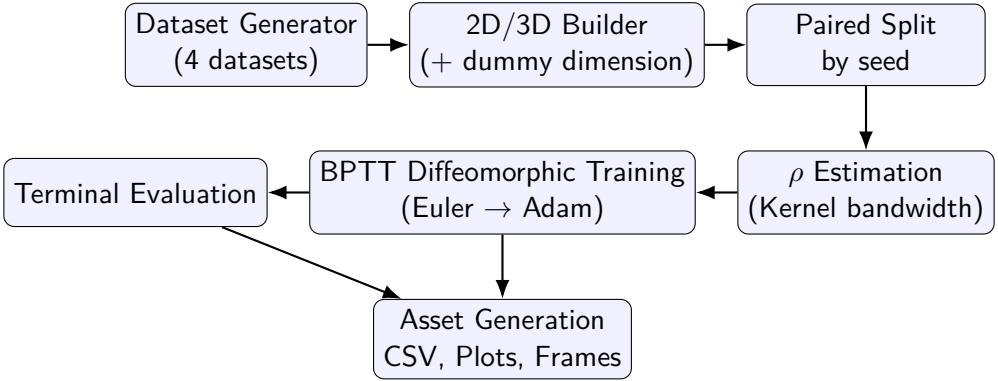


Maximum performance climb (+7.2% Delta): Complete spatial unlocking using the hidden vertical axis yielding a smooth, continuous topology lift over intersecting sets.

Implementation Schema



Conclusion

- **Topological Flow:** Diffeomorphic learning is a physically intuitive optimal transport model replacing the "black-box" mechanics of standard NNs.
- **The Core Tricks:** Incorporating Affine extensions breaks invariance, while adding a single Dummy coordinate completely solves the restriction of topological preservation.
- **Numerical reality:** The $O(N^2)$ explicit calculation is a heavy bottleneck; future work must concentrate on sparse parametric approximations rather than exact numerical representation.

Thank you.